Precalculus Review for Calculus

Section 1: Functions, Domain, and Composition

1 Given $f(x) = \sqrt{x - 3}$ and $g(x) = x^2 - 1$, find the domain of $(f \circ g)(x)$.

a. $(-\infty, -2] \cup [2, \infty)$

b. $[2, \infty)$

c. $[3, \infty)$

d. $[-2, 2]$

2 If $f(x) = \frac{2x+1}{x-3}$, what is $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{x-2}$

b. $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$

d. $f^{-1}(x) = \frac{2x-1}{x+3}$

3 Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 3x^2 - 5x$.

a. $6x + 3h - 5$

b. $6x - 5$

c. $3x + 3h - 5$

d. $6x + h - 5$

Section 2: Exponentials, Logarithms, and Trig Review

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4 Solve the equation $4^{x+1} = 8^{2x-1}$ for x.

a. $5/4$

b. 1

c. $3/2$

d. $2/3$

5 Solve $\ln(2x-1) + \ln(x) = \ln(3)$.

a. $x = 3/2$

b. $x = 1$

c. $x = 2$

d. $x = 3$

6 Find the exact value of $\sin\left(\frac{7\pi}{6}\right) + \cos\left(\frac{4\pi}{3}\right)$.

a. -1

b. $-1/2$

c. 0

d. $-\sqrt{3}$

Section 3: Rational Functions, Holes, and Asymptotes

7 Identify the vertical and horizontal asymptotes of $f(x) = \frac{3x^2 - 12}{x^2 - 5x + 6}$.

a. Vertical: $x = 3$; Horizontal: $y = 3$

b. Vertical: $x = 2, x = 3$; Horizontal: $y = 3$

c. Vertical: $x = 3$; Horizontal: $y = 0$

d. Vertical: $x = 2$; Horizontal: $y = 1$

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8 Determine the slant asymptote of $g(x) = \frac{2x^2 + 5x - 1}{x + 1}$.

a. $y = 2x + 3$

b. $y = 2x + 5$

c. $y = 2x - 1$

d. $y = x + 3$

9 Find the domain of $h(x) = \frac{\ln(x + 2)}{\sqrt{9 - x^2}}$.

a. $(-2, 3)$

b. $[-2, 3]$

c. $(-3, 3)$

d. $(0, 3)$

Section 4: Trigonometric Identities and Equations

10 Simplify $\frac{\sin(2\theta)}{1 + \cos(2\theta)}$ into a single trigonometric function.

a. $\tan\theta$

b. $\cot\theta$

c. $\sec\theta$

d. $\sin\theta$

11 Solve $2\cos^2 x + \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

a. $\pi/3, \pi, 5\pi/3$

b. $\pi/6, \pi, 11\pi/6$

c. $\pi/3, 2\pi/3$

d. $\pi/2, 3\pi/2$

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12 If $\sin\alpha = 3/5$ with $\alpha \in (\pi/2, \pi)$, find $\cos(2\alpha)$.

a. $7/25$

b. $-7/25$

c. $24/25$

d. $-24/25$

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